



# TEST REPORT

Part NO : SCB-3750/13.8

Rated Power : 3750 kVA

Series No. : MYG2505324

Finish Date : 2025/7/29

Check : Shen qian

Approve : Xu yangjie

JZPE POWER TRANSFORMER(JIANGSU)CO.,LTD.

Add: Longxi Industrial Zone,JiangYan District,Taizhou City,Jiangsu Province China

Tel: 0523-88601688

web: [www.jiezoupower.com](http://www.jiezoupower.com)



## TEST REPORT

p- 1

Manuf. No. MYG2505324

- |                               |                      |
|-------------------------------|----------------------|
| 1) Part Name:                 | Dry type transformer |
| 2) Series No.:                | MYG2505324           |
| 3) Type:                      | SCB-3750/13.8        |
| 4) Rated Power:               | 3750kVA              |
| 5) Rated Current:             | 156.89/3608.54 A     |
| 6) Rated Voltage:             | 13.8±2X2.5% / 0.6kV  |
| 7) Phase:                     | Three Phase          |
| 8) Rated Frequency:           | 60Hz                 |
| 9) Vector Group:              | Dyn1                 |
| 10) Cooling Method:           | AN                   |
| 12) Manufacture Date:         | 2025/7/29            |
| 13) Insulation Grade:         | 220°C                |
| 14) Applicable Condition:     | Indoor               |
| 15) Total Weight:             | 7870kg               |
| 16) Altitude above sea level: | ≤1000m               |
| 17) Standard of Test.         | CSA C88-16           |



# TEST REPORT

p- 2

Manuf. No. MYG2505324

## 1. DC Resistance Test.

Temperature: 33.0℃ Humidity: 55.0%

### HV Winding (Ω)

| Test position | Design value | H1H2    | H2H3   | H3H1   | (MAX-MIN)/<br>AVERAGE(%) | ≤2 |
|---------------|--------------|---------|--------|--------|--------------------------|----|
| 14490V        | 1.7070       | 0.2710  | 0.2710 | 0.2711 | 0.04                     |    |
| 14145V        | 1.6664       | 0.2644  | 0.2645 | 0.2646 | 0.08                     |    |
| 13800V        | 1.6257       | 0.2590  | 0.2580 | 0.2587 | 0.39                     |    |
| 13455V        | 1.5851       | 0.2520  | 0.2523 | 0.2525 | 0.20                     |    |
| 13110V        | 1.5444       | 0.245.5 | 0.2456 | 0.2460 | 0.16                     |    |

### LV Winding (mΩ)

| Test position | Design value | X1X2      | X2X3        | X3X1   | (MAX-MIN)/<br>AVERAGE(%) |
|---------------|--------------|-----------|-------------|--------|--------------------------|
| 600V          | 0.2697       | 0.3012    | 0.2941      | 0.3011 | 2.38                     |
| Test position | Design value | X1-X0     | X2-X0       | X3-X0  | (MAX-MIN)/<br>AVERAGE(%) |
| 600V          | 0.1348       | 0.1438    | 0.1442      | 0.1438 | 0.28                     |
| Conclusion    | OK           | Tested by | Xu Yang Jie | Date   | 2025/7/29                |

## 2 Voltage ratio Test.

Temperature: 33.0℃ Humidity: 55.0%

| Test position |           | Vector Group   | Dyn1      |             |           | Maximum deviation<br>( %) |      |
|---------------|-----------|----------------|-----------|-------------|-----------|---------------------------|------|
|               |           | Standard ratio | H1H2/X1X2 | H2H3/X2X3   | H3H1/X3X1 |                           |      |
| 1             | 14490/600 | 24.15          | 24.13     | 24.13       | 24.14     | -0.07                     | ±0.5 |
| 2             | 14145/600 | 23.58          | 23.56     | 23.56       | 23.56     | -0.08                     |      |
| 3             | 13800/600 | 23.00          | 23.04     | 23.04       | 23.04     | 0.19                      |      |
| 4             | 13455/600 | 22.43          | 22.47     | 22.47       | 22.47     | 0.19                      |      |
| 5             | 13110/600 | 21.85          | 21.89     | 21.89       | 21.89     | 0.18                      |      |
| Conclusion    |           | OK             | Tested by | Xu Yang Jie | Date      | 2025/7/29                 |      |



# TEST REPORT

p- 3

Manuf. No. MYG2505324

## 3. Insulation Resistance Test.

Temperature: 33.0℃

Humidity: 55.0%

| Test position    |    | Test results(MΩ)      | Standard of judgement | Test equipment               |           |
|------------------|----|-----------------------|-----------------------|------------------------------|-----------|
| HV----LV & Earth |    | 2500                  | ≥ 1000 MΩ             | 2500V/2500MΩ<br>5000V/5000MΩ |           |
| LV---HV & Earth  |    | 2500                  | ≥ 1000 MΩ             |                              |           |
| Conclusion       | OK | Tested by Xu Yang Jie |                       | Date                         | 2025/7/29 |

## 4. Applied Voltage Withstand Test.

Temperature: 33.0℃

Humidity: 55.0%

|                  |    |                  |           |             |  |              |  |
|------------------|----|------------------|-----------|-------------|--|--------------|--|
| Test position    |    | Test voltage(kV) |           | Current(mA) |  | Test time(s) |  |
| HV----LV & Earth |    | 34               |           | 24.70       |  | 60           |  |
| LV---HV & Earth  |    | 4                |           | 7.31        |  | 60           |  |
| Conclusion       | OK | Tested by        | Shen qian | Date        |  | 2025/7/29    |  |

## 5. Induced Voltage Withstand Test.

Temperature: 33.0℃

Humidity: 55.0%

| Test position |    | Test Voltage(V) |           | Frequency (Hz) |      | Test Time (s) |
|---------------|----|-----------------|-----------|----------------|------|---------------|
| 600V          |    | 1200            |           | 120            |      | 60            |
| Conclusion    | OK |                 | Tested by | Shen qian      | Date | 2025/7/29     |

## 6. Short-circuit Impedance and Load Loss.

Temperature: 33.0℃

Humidity: 55.0%

| Test position | Test Voltage (V) | Test current (A) | Test loss (W) | %I <sub>Z</sub> | %I <sub>R</sub> | %I <sub>X</sub> | 85℃             |                 |          |            |
|---------------|------------------|------------------|---------------|-----------------|-----------------|-----------------|-----------------|-----------------|----------|------------|
|               |                  |                  |               |                 |                 |                 | %I <sub>Z</sub> | %I <sub>R</sub> | loss (W) | Efficiency |
| 13800V        | 1255             | 156.9            | 19604         | 9.10            | 0.52            | 9.08            | 9.10            | 0.59            | 21997    | 99.50      |
| Conclusion    | OK               | Tested by        | Shen qian     | Date            | 2025/7/29       |                 |                 |                 |          |            |

## 7. Test on no-load current and no-load loss.

Temperature: 33.0℃

Humidity: 55.0%

| Test voltage(V) |     | No-load current (A) | No-load loss (60Hz) | No-load current | Conclusion |
|-----------------|-----|---------------------|---------------------|-----------------|------------|
| 1.0U            | 600 | 7.070               | 3838                | 0.20            | OK         |
| Conclusion      | OK  | Tested by           | Shen qian           | Date            | 2025/7/29  |



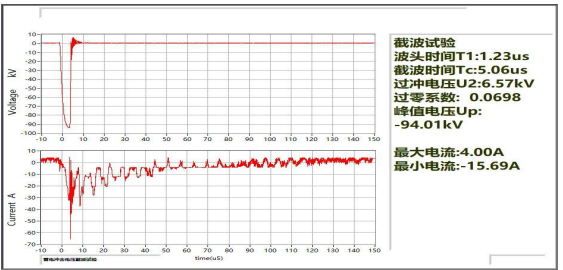
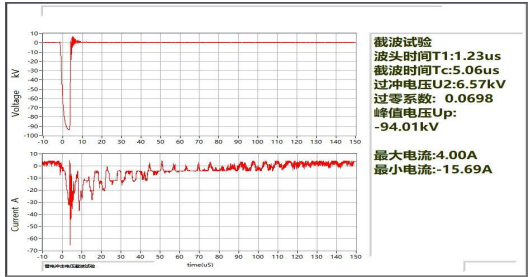
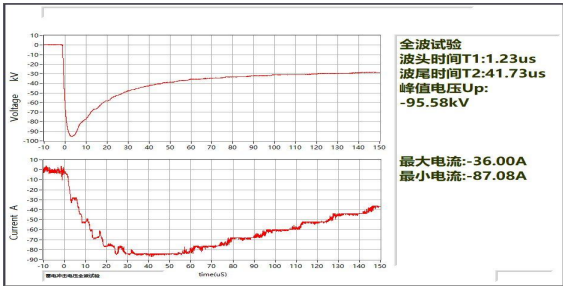
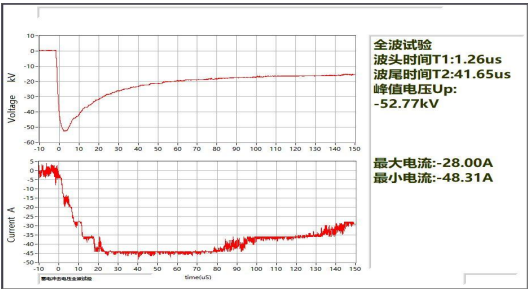
Ambient Temp.:33℃  
Relative Humidity: 57%  
Oil Temp. Indicator: 33℃

|   |         |         |      |       |            |
|---|---------|---------|------|-------|------------|
| 1 | H2 Tap3 | 50%FW   | 47.5 | 49.18 | 1.22/40.18 |
| 2 | H2 Tap3 | 100% FW | 95   | 95.61 | 1.21/40.39 |
| 3 | H2 Tap3 | 100% CW | 95   | 92.18 | 1.13/5.42  |
| 4 | H2 Tap3 | 100% CW | 95   | 94.4  | 1.20/4.94  |

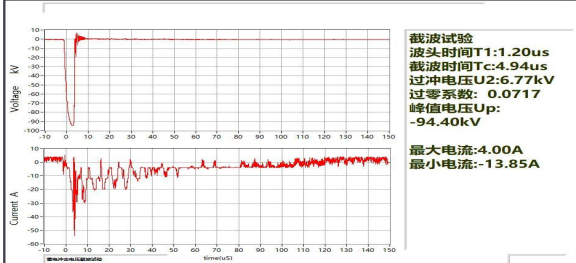
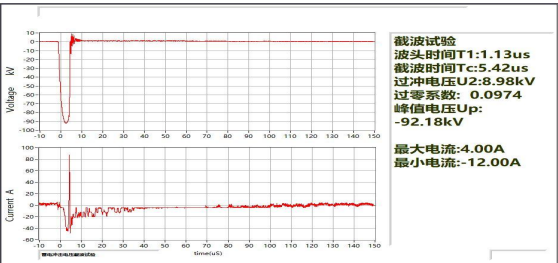
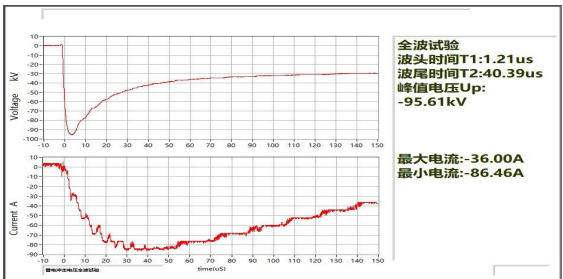
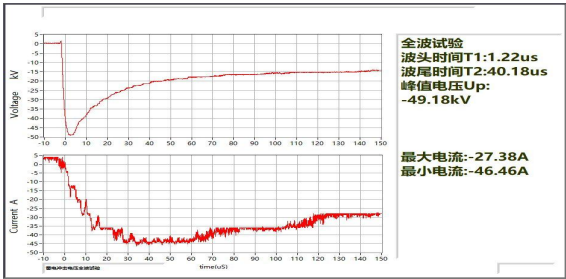
[illegible]

|   |         |         |      |       |            |
|---|---------|---------|------|-------|------------|
| 1 | H3 Tap5 | 50%FW   | 47.5 | 50.38 | 1.25/40.42 |
| 2 | H3 Tap5 | 100% FW | 95   | 95.31 | 1.21/42.89 |
| 3 | H3 Tap5 | 100% CW | 95   | 95.03 | 1.24/4.78  |
| 4 | H3 Tap5 | 100% CW | 95   | 94.65 | 1.21/4.72  |

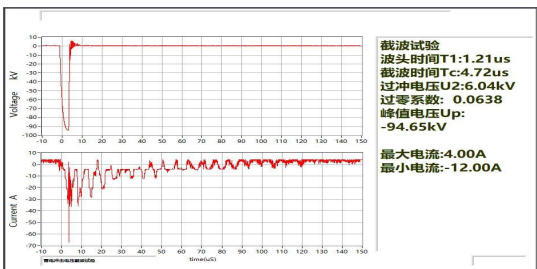
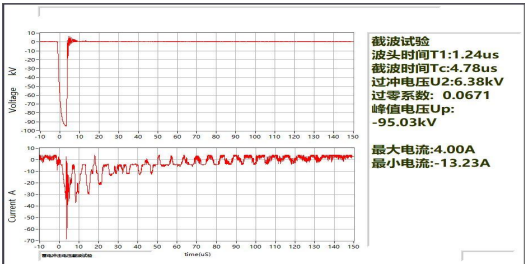
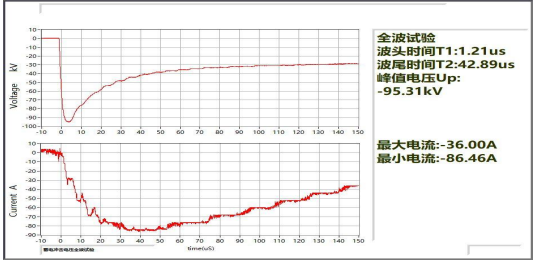
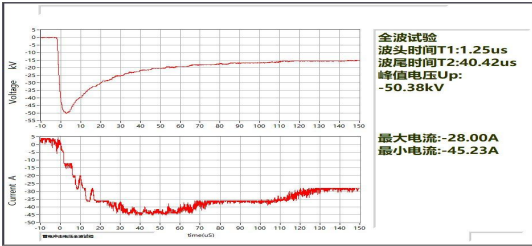
Applied Terminal:: H1  
Polarity: Negative  
Channels : Voltage wave



Applied Terminal:: H2  
Polarity: Negative  
Channels : Voltage wave



Applied Terminal:: H3  
Polarity: Negative  
Channels : Voltage wave



Criterion:

- 1.CW:Chopping wave( HV winding flashover time 2.0~6.0μs, LV winding flashover time min 1.0μs )  
2.FW: Full wave (front time T1=1.20μs±30%,(last time T2=50.0μs±20%)

Interpretation of Result:

All of the tested data comply with criterion

The test results meet the requirements of CSA C88 series standard and manufacture's technical specification.

Date: 2025/7/29

Test company: JZPE POWER TRANSFORMER(JIANGSU)CO.,LTD.

Add:Longxi Industrial Zone,JiangYan District,Taizhou City,Jiangsu Province China

[www.jiezoupower.com](http://www.jiezoupower.com)

# Temperature Rise detection Report

## Product specification

|                   |                      |                       |             |                     |              |
|-------------------|----------------------|-----------------------|-------------|---------------------|--------------|
| Model             | SGB-3750/13.8        | Vector group          | Dyn1        | Series No           | MYG2505325   |
| Rated capacity    | 3750KVA              | Insulation level      | 220℃        | Product code        | 1RZ.710.C812 |
| Service Condition | Indoor type          | Frequency             | 60Hz        | Cooling mode        | AN           |
| Rated parameter   | Total number of taps | High voltage winding. |             | Low voltage winding |              |
|                   |                      | Voltage (V)           | Current (A) | Voltage (V)         | Current (A)  |
|                   | 5                    | 13800                 | 165.15      | 600                 | 3608.5       |

Reference standard: CSA C88-16

Test time : 2025/7/29

## Test method

The transformer temperature rise test adopts the simulated load method, with the tap position at the 5 tap and no fan turned on.

No load temperature rise: Apply rated voltage of 600 V on the low voltage side.

Load temperature rise: A specified current of 165.15 A should be applied on the high voltage side, but the actual applied current is 164.08 A.

## Coil resistance

| Resistance measurement | HV coil            |                                  | LV coil            |                                  |
|------------------------|--------------------|----------------------------------|--------------------|----------------------------------|
|                        | Resistance RBC (Ω) | Measured ambient temperature (℃) | Resistance RBC (Ω) | Measured ambient temperature (℃) |
| Coil cold resistance   | 0.2513             | 34                               | 0.2981             | 34                               |

Measurement results under no-load loss

| winding                 | Measured ambient temperature (°C) | Winding temperature rise (K) |
|-------------------------|-----------------------------------|------------------------------|
| High-voltage winding(Ω) | 35.33                             | 2.01                         |
| Low-voltage winding(Ω)  | 51.89                             | 18.58                        |

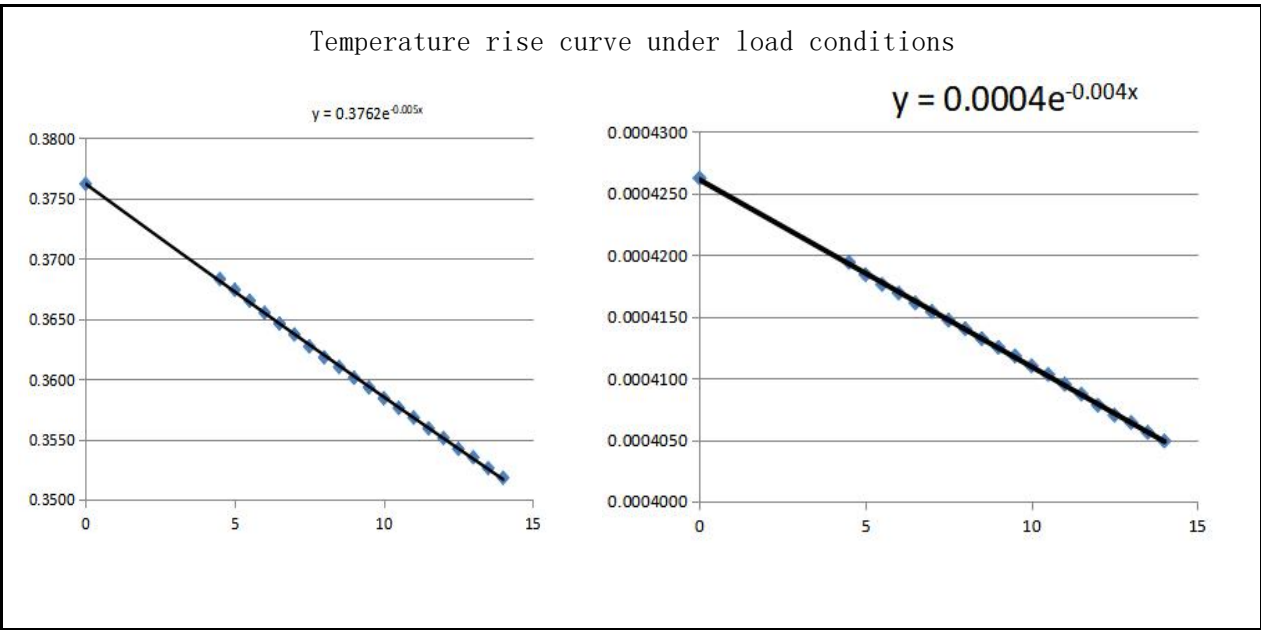
Measurement results under load Loss

| winding                 | Measured ambient temperature (°C) | Winding temperature rise (K) |
|-------------------------|-----------------------------------|------------------------------|
| High-voltage winding(Ω) | 176.31                            | 150.6                        |
| Low-voltage winding(Ω)  | 149.48                            | 122.95                       |

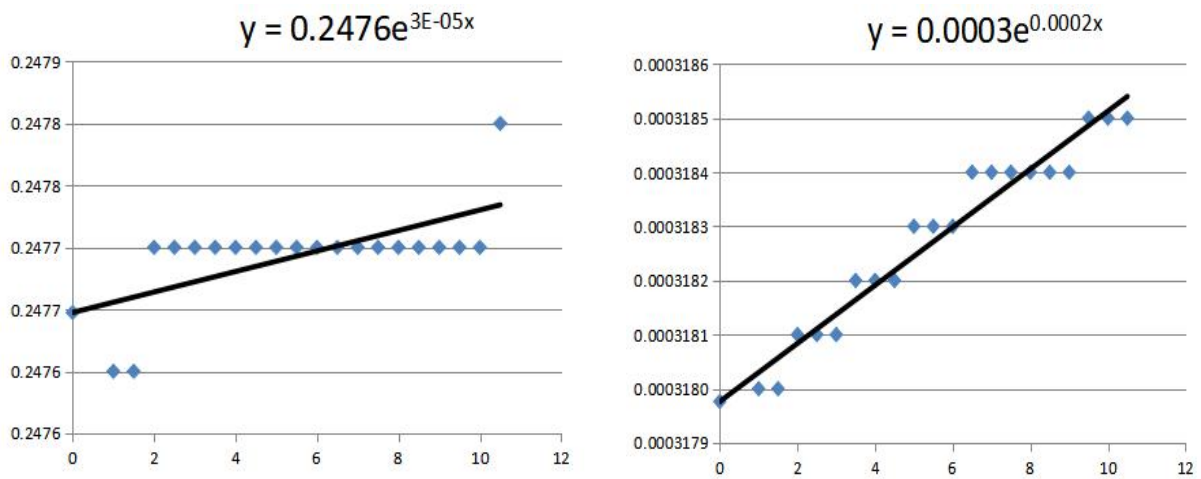
Total temperature rise of winding results

|                                   | High-voltage winding (K) | Low-voltage winding (K) |
|-----------------------------------|--------------------------|-------------------------|
| Standard value                    | 150                      | 150                     |
| Total temperature rise of winding | 147                      | 132.13                  |

Temperature rise Cooling curve diagram



Temperature rise curve under no-load conditions



Test conclusion: This test method and data meet CSA C88-16 requirements, the test is qualified

Date:2025/7/29